SAVEETHA SCHOOL OF ENGINEERING

OPERATING SYSTEMS

ASSIGNMENT | CO1

EVALUATE AND JUSTIFY THE OPERATING-SYSTEM DESIGN OF A SMART HOSPITAL

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Course	CSA0421 – Operating Systems
Assignment	Assignment 1 (Tasks 12)

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT THE DESIGN

Changed requirement: the hospital now supports remote patient-monitoring devices in patients' homes. These devices send readings through the internet. Patient records are moved to a cloud-based service shared with the partner hospitals. The original design must therefore be evaluated for a wider, less controlled network and a distributed data platform.

A. Impact analysis

The bedside monitor type is extended to a home-monitoring device. It still needs reliable sampling, local threshold detection and a local alert. If the device is battery-powered, mobile/embedded power management becomes more important. A cloud connection does not replace local real-time behaviour because internet latency and outages are not bounded to 10 ms.

Affected area	Impact of new requirement
OS type	Home devices need a small embedded real-time OS or a secure embedded Linux configuration according to hardware capacity and deadline. The cloud record service uses server OS instances in a distributed/cloud environment.
OS services	Secure network communication, buffering, retry, device identity, remote update, logging, time synchronization and health monitoring become essential on home devices.
Protection	Devices need unique identities, secure boot where supported, signed updates, encrypted communication, key rotation, least-privilege credentials and protection against unauthorized pairing.
Data management	The cloud service needs authenticated APIs, tenant/site separation, access policies, replication, backup, audit logging, data retention and disaster recovery.
Network assumptions	Internet connectivity is intermittent and has variable latency. Data must be queued locally and delivered safely after reconnection.

B. Evaluation of the original design

The original design remains correct in its most important principle: the bedside alarm is local and must not depend on a remote server. Its local RTOS and local alarm path can also be used in a home device if hardware and sensor drivers are suitable. However, the original hospital-only network and local database cluster are not sufficient by themselves for home-to-cloud communication and shared records across three hospitals.

Response time: the device can detect and alarm locally within its engineered deadline, but internet delivery cannot be guaranteed to arrive within 10 ms. The remote clinician notification is a secondary best-effort or service-level objective, not the primary safety alarm. Reliability: a home device must buffer readings during network outage and send them later. Security: devices outside the hospital perimeter are exposed to physical access, insecure Wi-Fi and internet threats, so stronger device identity, encryption and secure updates are required.

C. Adapted design – MODIFY the original plan

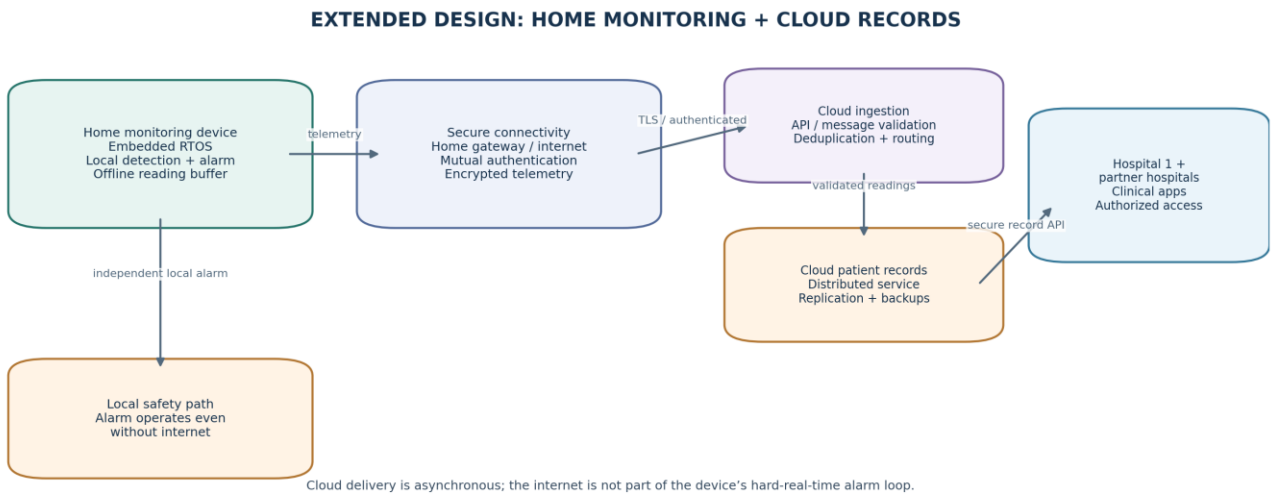


Figure 8. Home devices keep local alarm handling while readings are securely delivered to distributed cloud records for authorized hospital access.

The recommended decision is MODIFY, not fully redesign. Retain the local hard-real-time design for bedside monitors and apply the same local safety principle to home-monitoring devices. Add the following components:

- Home device layer: a small embedded RTOS or appropriately hardened embedded OS performs sensor acquisition, threshold checks, local alerting and local buffering. The primary safety alarm remains available without internet access.
- Secure home gateway / connectivity: devices authenticate to the service using unique credentials and mutually authenticated encrypted communication. Readings carry device identity, timestamp, sequence number and integrity information.
- Cloud ingestion service: an authenticated API or message-ingestion service validates device identity and message format, applies rate limits, detects duplicates and routes data to the appropriate patient record.
- Cloud patient-record service: distributed server instances provide shared record access across the three hospitals. Use replication, backups, monitoring, encryption, role-based access, audit logs and region/zone resilience appropriate to the service requirements.
- Hospital integration: clinical applications access cloud records through secure interfaces. Local caches may retain essential read-only information for downtime, with controlled reconciliation when connectivity returns.
- Operations and security: maintain an asset inventory, device revocation process, signed firmware updates, key rotation, vulnerability response, backup restore tests and incident monitoring.

D. Evidence-based defence and trade-offs

Decision	Benefit	Trade-off / control
Retain local real-time alarm	Preserves bounded alarm response even during WAN outage.	Requires device-level validation and local audible/visual alert mechanisms.
Add cloud shared records	Enables authorized clinicians at three hospitals to access common patient information and scales storage/compute.	Introduces WAN dependency, cloud-provider dependency and complex identity/data governance; use redundancy, backups and a downtime mode.

Decision	Benefit	Trade-off / control
Buffer readings at home	Prevents temporary network loss from discarding observations.	Requires bounded storage, clear overflow policy, timestamps and sequence numbers to detect gaps and duplicates.
Encrypt and authenticate device traffic	Protects confidentiality, integrity and device identity over untrusted networks.	Requires credential provisioning, certificate/key rotation and recovery when a device is lost or compromised.
Use distributed record architecture	Supports multiple sites and cloud scale.	Consistency, replication lag, conflicts and network partition behaviour must be explicitly designed and tested.

A safe data-flow rule is: detect and alarm locally first; transmit readings asynchronously; acknowledge durable receipt at the cloud service; and retain the local copy until a defined delivery confirmation is received. Sequence numbers and timestamps help the cloud service identify missing, repeated or out-of-order readings. If a device is offline, it should not falsely report successful cloud delivery.

Final defence: MODIFY the original design. Keep dedicated real-time processing at the patient device, while extending the data and record layers into a secure distributed cloud service. This preserves the original 10 ms safety objective and supports remote monitoring and record sharing without assuming that the public internet is deterministic.